Pars plana vitrectomy to repair retinal detachment following brachytherapy for uveal melanoma

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ABSTRACT

Aims Retinal detachment can develop following brachytherapy for uveal melanoma. This complication may result in substantial visual loss and poses a significant therapeutic dilemma due to the required surgical intervention for correction of the detachment. We report the incidence of retinal detachment in eyes treated with brachytherapy for posterior uveal melanoma and the outcome of pars plana vitrectomy in those eyes.

Methods Patients who developed tractional or combined tractional–rhegmatogenous retinal detachment following brachytherapy for posterior uveal melanoma in a single referral centre were retrospectively evaluated. Clinical findings, demographics, and ophthalmic imaging findings were recorded, as well as the manner of treatment and its success.

Results Seven of the 473 posterior uveal melanoma patients (1.48%), who were treated between 2000 and 2011 with brachytherapy, developed tractional or combined tractional–rhegmatogenous retinal detachment. Retinal detachment developed at a mean of 50.1 months (range 3.5–120 months) following brachytherapy. All patients underwent pars plana vitrectomy. Retinas remained attached in each of the cases. In five of seven patients there was substantial improvement in visual acuity following repair of the retinal detachment. No tumour growth or dissemination were observed during the mean follow-up of 18.4 months after vitrectomy (range 10–36 months).

Conclusions Tractional and tractional–rhegmatogenous retinal detachment are rare complications of treated uveal melanoma. Pars plana vitrectomy appears to be an effective and safe procedure in such cases.

INTRODUCTION

Brachytherapy is commonly used to treat uveal melanoma, and usually achieves control of local tumour growth.[1–7] Successful brachytherapy may minimise visual loss due to tumour growth and can prevent the visual loss, psychological consequences, and cosmetic blemish associated with enucleation. Brachytherapy leads to tumour cell necrosis, vascular obstruction, and fibrosis of the tumour stroma with accumulation of pigmented macrophages and subretinal gliosis, as well as to chorioretinal atrophy and scarring of the sclera within the field of radiation.[8] These processes might also lead to the development of exudative retinal detachment and sub- or epiretinal fibrosis which can be associated with tractional and/or rhegmatogenous retinal detachment.

Tumour spread is a concern when contemplating intraocular surgery in eyes harbouring uveal melanoma. Yet studies suggest that cataract extraction from eyes with irradiated uveal melanoma does not affect metastasis or mortality rates.[9–11] While cataract surgery does not require entry into the vitreous or manipulation of the retina in proximity to the melanoma, pars plana vitrectomy for correction of retinal detachment following brachytherapy may interact with the residual tumour. However, as surgery is done after adequate plaque radiation has been performed, the risk for tumour growth or spread seems to be low.

There are a few reports of pars plana vitrectomy following brachytherapy for uveal melanoma, and the most common indication for such intervention is vitreous haemorrhage.[12–14] Foster *et al*[13] described nine patients who underwent vitrectomy after plaque radiotherapy for a vitreoretinal abnormality, most due to vitreous haemorrhage. Bianciotto and colleagues reported on 74 patients who underwent vitrectomy for vitreal haemorrhage removal after tumour regression.[12] Haimovici *et al*[14] described six patients in whom rhegmatogenous retinal detachment developed after radiotherapy. Retinal detachment was repaired successfully in five of the six patients with scleral buckling alone, or in combination with pars plana vitrectomy in one patient.

Retinal detachment often results in additional visual loss in eyes where some useful vision was preserved following brachytherapy. In addition to the concern related to tumour spread and bleeding from the remaining tumour tissue, repair of retinal detachment in such eyes poses a technical challenge due the proximity of the membranes, bands, and retinal holes to the tumour. In this study, our aim was to determine the prevalence of this complication as well as the outcome and safety of pars plana vitrectomy for repair of retinal detachment in such eyes.

PATIENTS AND METHODS

A single-centre, retrospective, consecutive case series from a national specialised ocular oncology centre is described. Patients who underwent ruthenium-106 plaque brachytherapy treatment for posterior uveal melanoma between January 2000 and December 2011 were identified by computer search of the clinic database. From this search we found seven patients who underwent pars plana vitrectomy for rhegmatogenous or tractional retinal detachment which developed after the brachytherapy treatment. The incidence of retinal detachment in eyes treated with brachytherapy for posterior uveal melanoma was calculated. Baseline data recorded for each patient included patient age, sex, involved eye, tumour dimensions estimated by B-scan and A-scan echography at diagnosis, dates of brachytherapy and vitrectomy, and surgical technique utilised to repair the retinal detachment. Recorded follow-up data included the date of last

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follow-up, visual acuity, and ocular and systemic status at last follow-up. Approval of the institutional review board was obtained.

RESULTS

Between January 2000 and December 2011, 473 patients were treated with brachytherapy for posterior uveal melanoma in our medical centre. All patients were treated with a ruthenium-106 ophthalmic applicator. Seven of the 473 patients (1.48%) developed tractional or tractional–rhegmatogenous retinal detachment during follow-up. In six patients, a good clinical response to brachytherapy and stability during follow-up were demonstrated by their clinical appearance and ultrasound imaging characteristics. In another patient (patient 7 in table 1), an initial good clinical response to brachytherapy and stability were demonstrated during 19 months of follow-up, until an increase of tumour height (from 3.1 to 7.3 mm) and vitreous haemorrhage were detected. The patient was treated by stereotactic radiosurgery, after which the tumour regressed to 3 mm in height and the vitreous haemorrhage spontaneously resolved. The tumour remained stable until retinal detachment development 14 months later on, and also during the consecutive 10 months after vitrectomy. None of the patients had prior history of retinal holes or tears or retinal detachment in the treated or fellow eye.

Visual acuity at last visit before retinal detachment onset was 0.1 or better in each case, except for a patient with a macular hole (patient 3 in table 1) who had visual acuity of finger counting at 1.5 m. Two patients (patients 1 and 5 in table 1) had chronic, stable and small rimming exudative retinal detachment after brachytherapy for at least 6 months before total retinal detachment development. In both patients clinical appearance and ultrasound imaging characteristics of the tumour remained stable. At the time of diagnosis of retinal detachment, visual acuity was hand motion or finger counting in five patients with total retinal detachment and 0.2– or finger counting in two patients with partial retinal detachment and macula off or partially detached. Visual acuity decreased from a mean of 1.16 ±0.63 LogMAR following brachytherapy but before development of retinal detachment to a mean of 2.35±0.89 LogMAR at the time of retinal detachment diagnosis (p=0.032, t test).

Retinal detachment developed at a mean of 50.1 months (range 3.5–120 months) after the brachytherapy. Five patients developed total retinal detachment and two developed partial or sub-total retinal detachment. In six eyes at least one retinal break was detected. An atrophic hole was detected centrally, in proximity to the brachytherapy treated macular area in one eye. In five other eyes, one to three peripheral atrophic holes were detected which were removed from the tumour. A tractional retinal detachment component was demonstrated in four eyes. Patient data, initial tumour characteristics, treatment modalities, and treatment success in terms of visual acuity are summarised in table 1. Fundus photographs of patient 1 and 2 from table 1 are shown in figures 1 and 2, respectively.

In all seven patients uveal melanoma was stable without signs of tumour growth or systemic dissemination at the time of retinal detachment development and pars plana vitrectomy. Each patient underwent 23G vitrectomy, endolaser, and intravitreal silicone oil injection; six patients also underwent retinotomy and membrane peeling, two also underwent scleral buckling, and two underwent phacoemulsification with intraocular lens implantation during the same procedure. During the operation notable vitreoretinal adhesions were observed in six patients.

Table 1 Initial patient features and pretreatment tumour characteristics

Patient	Age	Gender	Eye	Initial basal tumour dimensions (mm)	BCVA at diagnosis of tumour	Initial tumour height (mm)	Interval from tumour treatment to PPV (months)	Type of RD	BCVA before RD	BCVA at diagnosis of RD	BCVA after RD repair	Time of F/U after PPV	Final tumour height (mm)
1	62	M	L	8.8×10.4	0.25	7	120	Total, TRD+RRD	0.1	HM	0.1	36	5.35
2	67	F	R	8.0×8.8	0.63	3.2	80	Total, TRD+RRD	0.5	HM	0.32	27	2.2
3	72	F	L	6.4×7.2	FC 0.3 m	2.56	3.5	Total, RRD	FC 1.5 m	HM	FC 0.3 m	18	2
4	56	M	R	9.3×9.4	0.16	7.7	39	Total, RRD	0.2–	HM	0.12	13.5	2
5	60	M	L	11.65×11.86	1.2	7.39	8.1	Macula off, TRD	0.2	0.2–	FC 1 m	12	6
6	68	M	L	11.2×13.2	0.16	5.6	74	Split macula, RRD	FC 2 m	FC 1.5 m	FC 1 m	12	3.94
7	65	M	R	15×16.44	1.5	9.11	33	Total, TRD+RRD	FC 2.5 m	FC 2 m	0.1–	10	3

BCVA, best corrected visual acuity; F/U, follow-up; F, female; FC, finger counting; HM, hand motion; L, left; M, male; PPV, pars plana vitrectomy; R, right; RD, retinal detachment; RRD, rhegmatogenous retinal detachment; TRD, tractional retinal detachment.

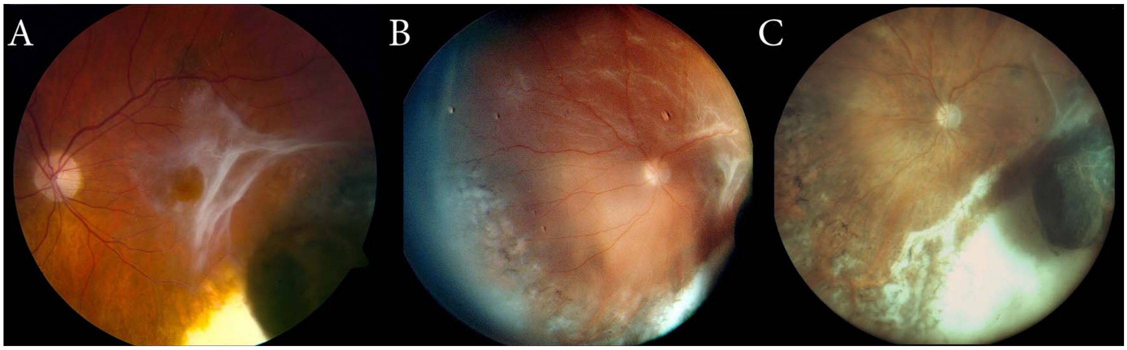

Figure 1 Colour fundus photographs of patient 1 (in table 1). (A) Eight years after brachytherapy, an epiretinal fibrotic membrane was detected in the proximity of the treated melanoma and overlying the macula. (B) Two years later the patient presented with decreased visual acuity, and total retinal detachment with atrophic retinal breaks at the edge of the tumour were observed. (C) Nine months following pars plana vitrectomy with retinotomy around the tumour, the retina was flat and a minor recurrence of epiretinal fibrotic membranes was evident; 28 months following the vitrectomy, the retina remained flat. Access the article online to view this figure in colour.

Follow-up lasted 10–36 months after retinal detachment repair. Substantial improvement in visual acuity was observed in each patient at the end of follow-up, except in patient 5 whose visual acuity deteriorated from 0.2– to finger counting, and in patient 6 whose visual acuity remained stable. In both patients the retina was flat and the eye was still filled with silicone oil at the end of follow-up. LogMAR visual acuity improved from a mean of 2.35 ± 0.89 before the surgery to 1.48 ± 0.79 at the end of follow-up (comparable for improvement from a mean of finger counting at 0.5 m to Snellen visual acuity of 20/600; $p=0.152$, t test). Silicone oil removal was carried out in three patients; in one patient an attempt to remove the silicone oil resulted in re-detachment during the surgery, the eye was refilled with silicone oil, and the retina was flat at the end of follow-up. No vitreous haemorrhage, tumour growth or intraocular or systemic dissemination were observed during the follow-up after vitrectomy.

DISCUSSION

Uveal melanoma is usually accompanied by a exudative retinal detachment at the time of diagnosis,[15] and this detachment usually resolves after brachytherapy.[16] It differs from late onset, post-brachytheraphy retinal detachment which is rare and mainly related to exudative detachment, subretinal fibrosis, retinal thinning, and atrophic retinal holes which can gradually develop in such eyes. Retinal breaks can develop secondary to the traction, particularly in areas of retinal atrophy which appear around the tumour following the brachytherapy.

In this article we present seven patients with brachytherapy-treated intraocular melanoma who underwent pars plana vitrectomy for repair of tractional retinal detachment or combined tractional–rhegmatogenous retinal detachment. In all seven patients the tumour has regressed and was stable before the development of retinal detachment. All patients had attached retinas at the end of follow-up; five of the seven patients gained visual acuity following the surgery, while two patients still had silicone oil in the eye which might be removed at a later date, possibly resulting in sub-optimal visual acuity.

Vitrectomy procedures in eyes with treated uveal melanoma carry potential hazards, among them tumour dissemination and intraocular bleeding. In a previous report of one case of pars plana vitrectomy for rhegmatogenous retinal detachment and vitreal haemorrhage in an eye with regressed uveal melanoma, no tumour recurrence was observed during the follow-up.[14] Foster *et al*[13] described nine patients who underwent vitrectomy after plaque radiotherapy for a vitreoretinal abnormality, most due to vitreous haemorrhage. Only one patient underwent vitrectomy for rhegmatogenous retinal detachment. There was no evidence of intraocular tumour dissemination in eight of the eyes. Only one patient, who had intratumoral and vitreous haemorrhage before plaque radiotherapy, developed intraocular tumour dissemination 56 months after vitrectomy. Bianciotto *et al*[12] reported on 74 patients who underwent vitrectomy for

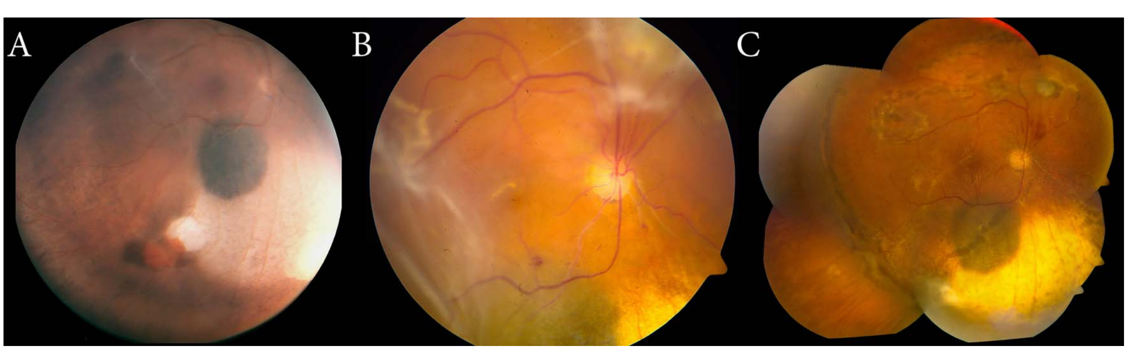

Figure 2 Colour fundus photographs of patient 2 (in table 1). (A) Five years after brachytherapy there is a fine subretinal membrane temporally to the macula and the retina is flat. (B) Six years and 8 months following the brachytherapy, the patient presented with reduced visual acuity and a total retinal detachment and two foci of subretinal fibrosis connected by subretinal band were detected. (C) Five months following pars plana vitrectomy and scleral buckle, the retina is flat, laser marks are seen superior to the vascular arcade, and an indentation of the scleral buckle is seen temporally. Access the article online to view this figure in colour.

vitreous haemorrhage which developed after brachytherapy that resulted in tumour regression. The incidences of local tumour recurrence, extraocular extension, and distant metastasis were similar to those in patients who did not have vitrectomy. Similarly, Damato *et al* reported on one patient with intrascleral tumour recurrence out of 52 tumour endoresection cases.[17 18]

Pars plana vitrectomy and manipulation of the retina, especially with epi- and subretinal membrane peeling and retinotomy, that were performed on some of our patients might potentially increase the risk of tumour reactivation or dissemination into the vitreous. Yet, in accordance with previous reports on intraocular surgery following brachytherapy, none of the patients had tumour growth or dissemination during the follow-up. Radiation therapy alters the tumour and induces fibrosis and necrosis of the tissue, as well as causing retinal fragility and thinning, resulting in a potentially increased risk for bleeding during vitrectomy. However, none of our patients had either intraoperative or postoperative bleeding.

In conclusion, according to our experience, pars plana vitrectomy for cases with tractional–rhegmatogenous retinal detachment in eyes treated with brachytherapy for posterior uveal melanoma is feasible and often results in anatomical and functional success. Longer follow-up and larger studies will be needed to evaluate the safety of vitrectomy for retinal detachment repair in eyes with treated uveal melanoma, comparing aged matched, sex matched vitrectomised patients versus non-vitrectomised patients with the same follow-up time from initial tumour diagnosis.

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Data sharing statement I have access to any data upon which the manuscript is based and will provide such data upon request to the editors or their assignees.

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